

Exercices on session *Tidyverse 1/2*

```
library(BIOMASS)
library(tidyverse)
```

For all the following exercises, use functions from the tidyverse whenever possible, and try to make one pipe per question.

Karnataka forest dataset

You are going to work on data from forests of the central Western Ghats of India. These data were available in a previous version of the package BIOMASS (v 2.1.11), and are called KarnatakaForest. You can access these data in a csv file [here](#).

- Read the data using the function read.csv (don't forget to look at the metadata in the help available here [here](#)).

```
dt_WG <- read.csv("data/KarnatakaForest.csv")
```

- Get an overview of the data and check their class.

```
str(dt_WG)
```

```
'data.frame':  65889 obs. of  8 variables:
 $ plotId : chr  "BSP1" "BSP1" "BSP1" "BSP1" ...
 $ treeId : chr  "BSP1_1" "BSP1_2" "BSP1_3" "BSP1_4" ...
 $ family : chr  "Fabaceae" "Fabaceae" "Fabaceae" "Fabaceae" ...
 $ genus  : chr  "Acacia" "Acacia" "Acacia" "Acacia" ...
 $ species: chr  "sinuata" "sinuata" "sinuata" "sinuata" ...
 $ D      : num  3.82 4.14 4.14 4.46 4.46 ...
 $ lat    : num  14.4 14.4 14.4 14.4 14.4 ...
 $ long   : num  74.9 74.9 74.9 74.9 74.9 ...
```

```
summary(dt_WG)
```

plotId		treeId		family		genus	
Length	:65889	Length	:65889	Length	:65889	Length	:65889
N.unique	: 96	N.unique	:65889	N.unique	: 73	N.unique	: 256
N.blank	: 0	N.blank	: 0	N.blank	: 0	N.blank	: 0
Min.nchar	: 4	Min.nchar	: 6	Min.nchar	: 7	Min.nchar	: 4
Max.nchar	: 6	Max.nchar	: 12	Max.nchar	: 17	Max.nchar	: 18
				NAs	: 60		

species		D		lat		long	
Length	:65889	Min.	: 3.183	Min.	:13.23	Min.	:74.15
N.unique	: 318	1st Qu.:	5.411	1st Qu.:	14.02	1st Qu.:	74.67
N.blank	: 0	Median	: 8.913	Median	:14.28	Median	:74.87
Min.nchar	: 3	Mean	: 14.493	Mean	:14.29	Mean	:74.89
Max.nchar	: 15	3rd Qu.:	18.144	3rd Qu.:	14.81	3rd Qu.:	75.12
NAs	: 62	Max.	:333.589	Max.	:15.24	Max.	:75.82

```
class(dt_WG)
```

```
[1] "data.frame"
```

- Transform the data to a tibble.

```
dt_WG <- as_tibble(dt_WG)
str(dt_WG)
```

```
tibble [65,889 x 8] (S3: tbl_df/tbl/data.frame)
 $ plotId : chr [1:65889] "BSP1" "BSP1" "BSP1" "BSP1" ...
 $ treeId : chr [1:65889] "BSP1_1" "BSP1_2" "BSP1_3" "BSP1_4" ...
 $ family : chr [1:65889] "Fabaceae" "Fabaceae" "Fabaceae" "Fabaceae" ...
 $ genus  : chr [1:65889] "Acacia" "Acacia" "Acacia" "Acacia" ...
 $ species: chr [1:65889] "sinuata" "sinuata" "sinuata" "sinuata" ...
 $ D      : num [1:65889] 3.82 4.14 4.14 4.46 4.46 ...
 $ lat    : num [1:65889] 14.4 14.4 14.4 14.4 14.4 ...
 $ long   : num [1:65889] 74.9 74.9 74.9 74.9 74.9 ...
```

- Now, read the data using the function `read_csv`. This is a preferred alternative as it returns directly a tibble

```
dt_WG <- read_csv("data/KarnatakaForest.csv")
```

- Transform all non-numeric variables to a factor.

```
dt_WG <- dt_WG %>%  
  mutate(plotId = as.factor(plotId),  
         treeId = as.factor(treeId),  
         family = as.factor(family),  
         genus = as.factor(genus),  
         species = as.factor(species))  
  
# alternative solution (not seen in the class, will be seen later)  
dt_WG <- dt_WG %>%  
  mutate(across(c(plotId, treeId, family, genus, species), as.factor))
```

Diameters

- Rename the column *D* to *diam*

```
dt_WG <- dt_WG %>% rename(diam = D)
```

- What is the smallest diameter of the whole dataset (make sure you extract the value, not only read it in a output)?

```
diam_min <- dt_WG %>%  
  slice_min(diam) %>% # get only the rows with the smallest diameter  
  select(diam) %>% # keep only the variable diameter  
  distinct() # keep only one unique value  
diam_min
```

```
# A tibble: 1 x 1  
  diam  
  <dbl>  
1  3.18
```

- What are the full species names (genus and species) of the 5 biggest trees?

```
big5 <- dt_WG %>%
  arrange(desc(diam)) %>% # sort the tibble by decreasing order of diameter
  slice_head(n=5) %>% # take the first 5 rows
  select(genus, species) # select the columns
big5
```

```
# A tibble: 5 x 2
  genus      species
  <fct>      <fct>
1 Ficus      virens
2 Dysoxylum malabaricum
3 Dysoxylum malabaricum
4 Dysoxylum malabaricum
5 Elaeocarpus tuberculatus
```

```
# alternative solution
big5 <- dt_WG %>%
  slice_max(diam, n=5) %>%
  select(genus, species)

# and to add a new colum for the full species name (work with both solutions)
dt_WG %>%
  slice_max(diam, n=5) %>%
  unite(full_sp,
        genus, species,
        sep = "_") %>%
  select(full_sp)
```

```
# A tibble: 5 x 1
  full_sp
  <chr>
1 Ficus_virens
2 Dysoxylum_malabaricum
3 Dysoxylum_malabaricum
4 Dysoxylum_malabaricum
5 Elaeocarpus_tuberculatus
```

- How many different species are amongst these five biggest trees?

```
big5 %>%
  distinct() %>% # select unique rows
  count() # count them
```

```
# A tibble: 1 x 1
      n
  <int>
1     3
```

```
# alternative solution that return a value and not a tibble
n_distinct(big5)
```

```
[1] 3
```

```
# NB: this is different from
# big5 %>% count(genus, species)
# which counts the number of tree for each species
```

- Create a table (tibble) with the median diameter per plot.

```
dt_med_diam <- dt_WG %>%
  group_by(plotId) %>% # group by plotId
  summarize(median_d = median(diam)) # take the median by plotId
dt_med_diam
```

```
# A tibble: 96 x 2
  plotId median_d
  <fct>     <dbl>
1 BSP1      6.37
2 BSP10     9.55
3 BSP100    6.05
4 BSP101    13.4
5 BSP102    14.6
6 BSP104     8.28
7 BSP11     8.59
8 BSP12     6.05
9 BSP13     9.55
10 BSP14    10.8
# i 86 more rows
```

- What is the most common family in the whole dataset?

```
com_fam <- dt_WG %>%
  count(family) %>% # count the number of trees per family
  slice_max(n) # get the one with the highest number of tree
com_fam
```

```
# A tibble: 1 x 2
  family      n
  <fct>    <int>
1 Combretaceae 8462
```

- Create a tibble with the number of trees per family and per plot for plots BSP1 and BSP2, and order it by decreasing number of individuals per family and per plot.

```
dt_sub <- dt_WG %>%
  filter(plotId %in% c("BSP1", "BSP2")) %>% # keep only the two plots of interest
  count(plotId, family) %>% # count the number of trees per family and plot
  arrange(plotId, desc(n)) # sort them
view(dt_sub)
```

- To how many families do trees that have a diameter in the fourth quartile (*i.e.* in the interval between the 3rd and the 4th quartile, *i.e.* bigger than the 3rd quartile) of the whole dataset belong?

NB: quartile can refer both to the value (most common use), or to an interval.

```
n_75 <- dt_WG %>%
  filter(diam > quantile(diam, probs = 0.75)) %>% # trees with diameter in the 4th quartile
  distinct(family) %>% # keep only a unique row for each family
  count() # count the rows
n_75
```

```
# A tibble: 1 x 1
      n
  <int>
1    60
```